

# Introduction to GIA

Lecture 1: A short history of GIA (Erik)

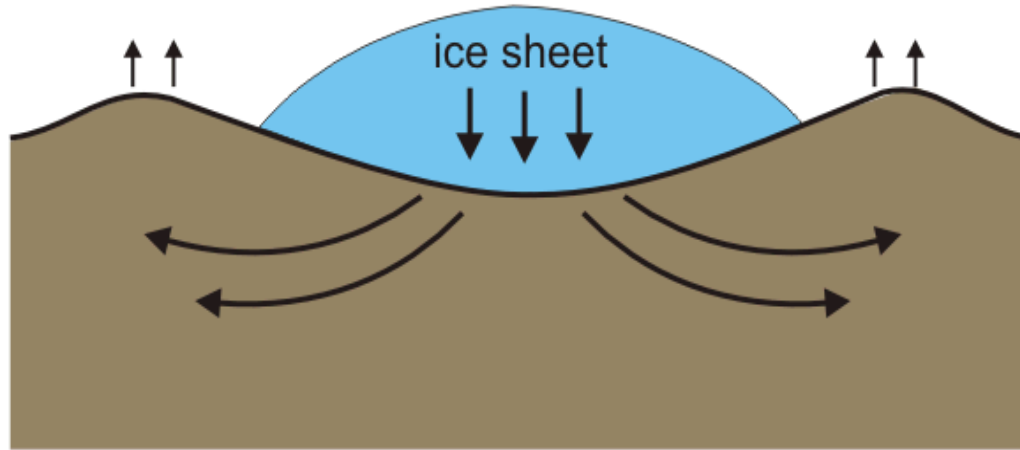
Lecture 2: Broad overview of GIA and model components (Glenn)

Lecture 3: Earth Rheology and Rotation (Erik)

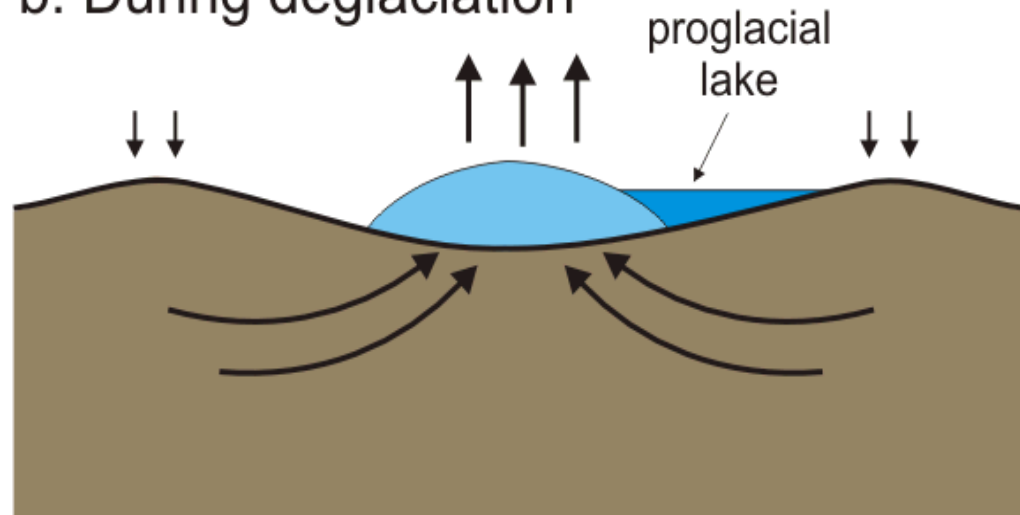
Lecture 4: GIA modelling and model applications (Glenn)

# Glacial Isostasy

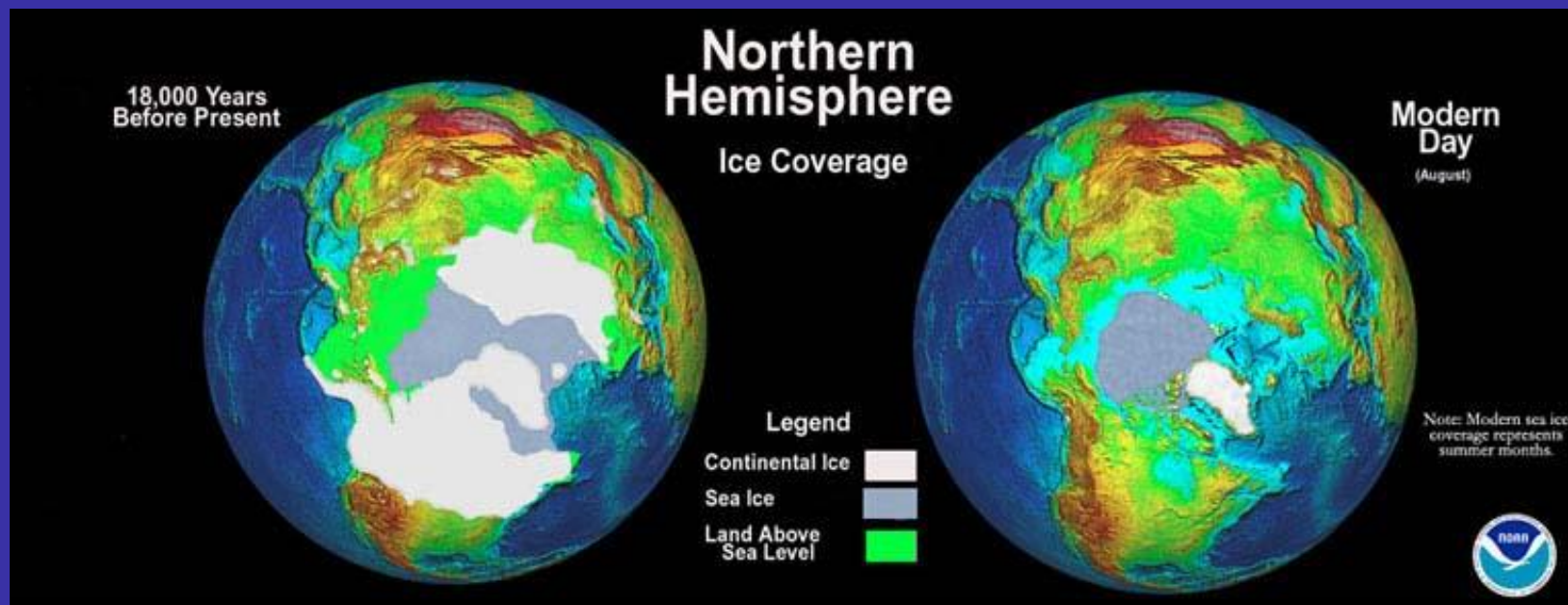
a. Peak glaciation



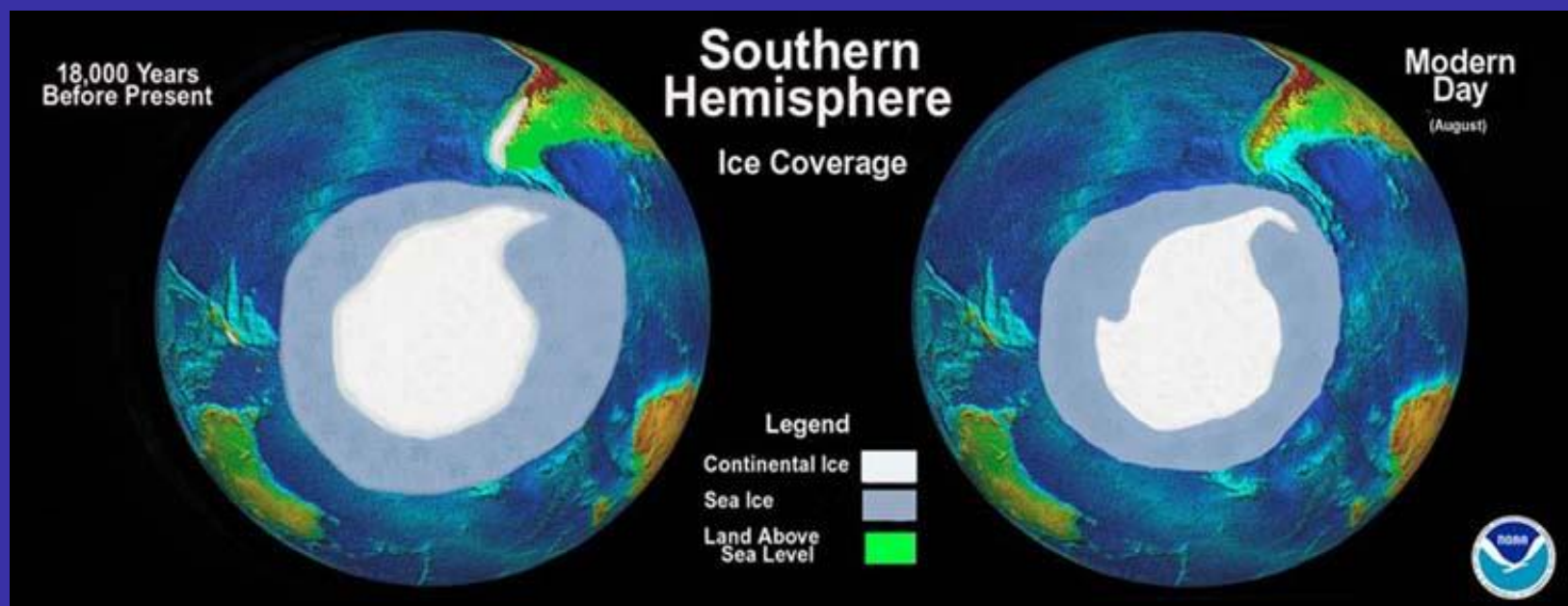
b. During deglaciation



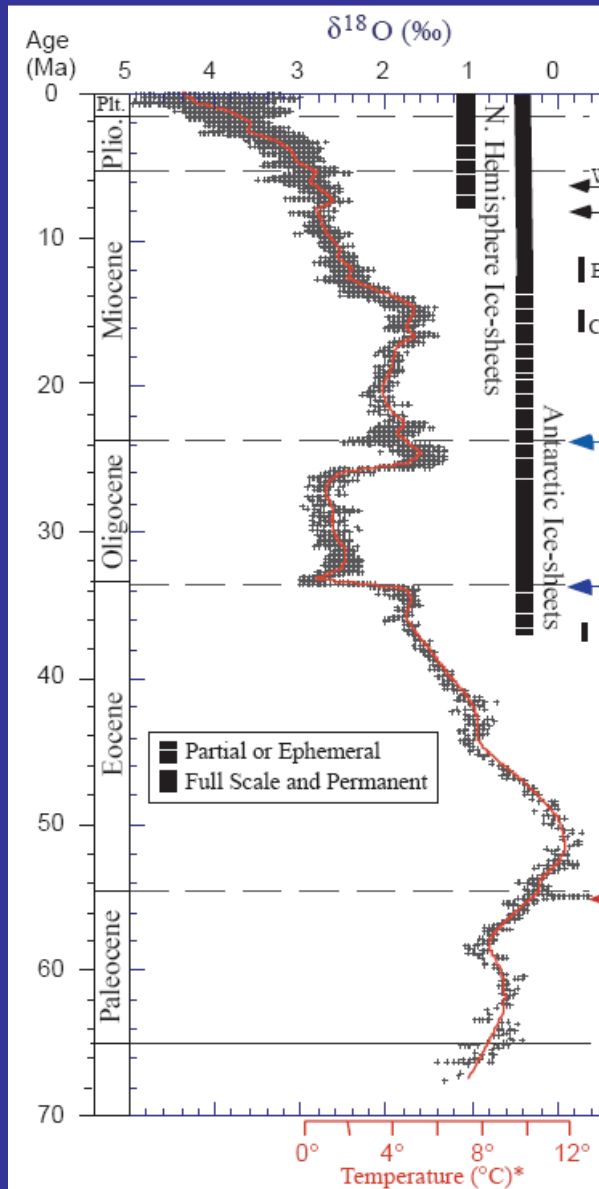
For a given loading history, the isostatic response is governed by the density and rheology of the Earth's interior.



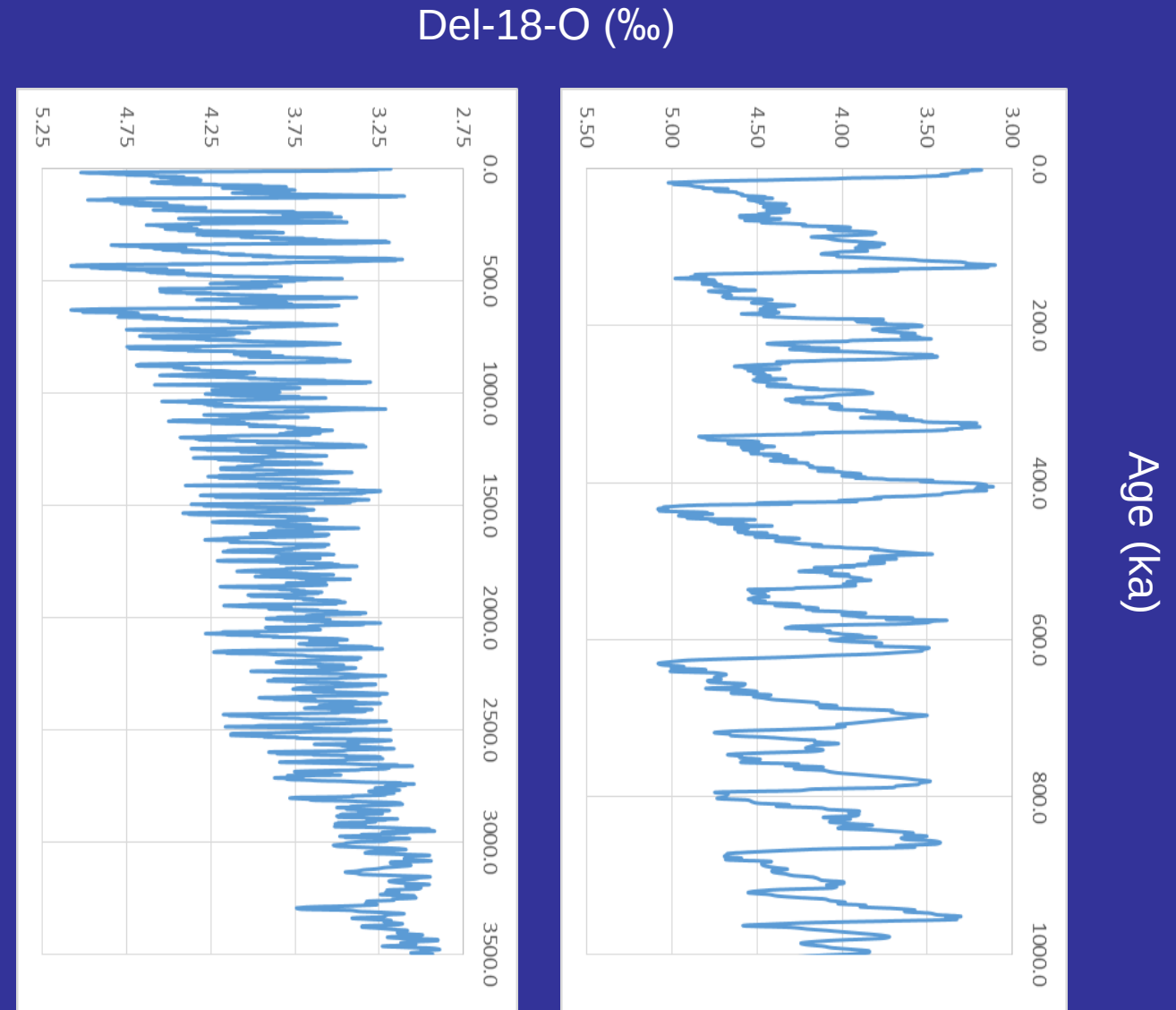
# Ice Extent at Last Glacial Maximum



# Time History of Ice: Marine Oxygen Isotope Data

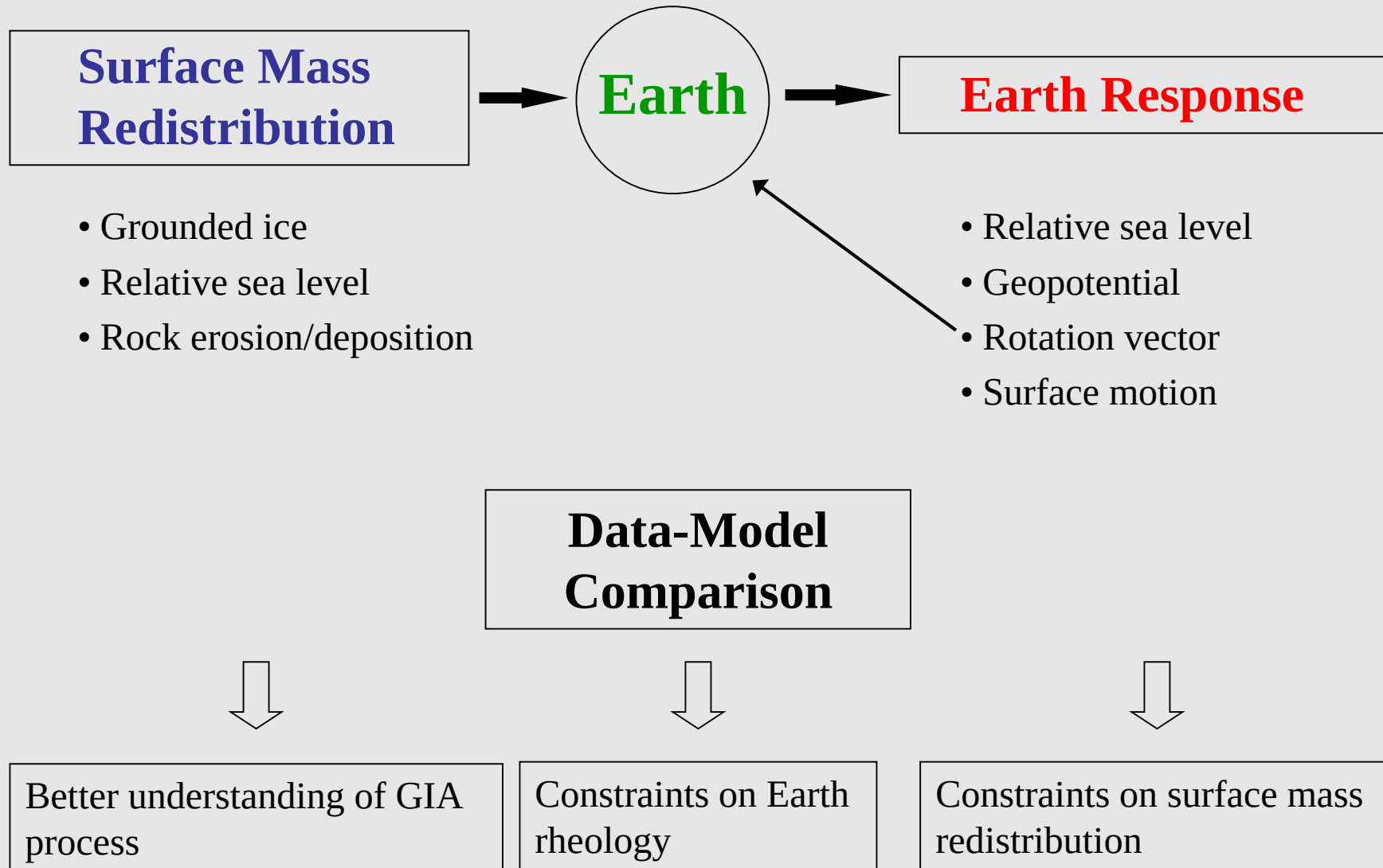


Zachos et al. (2001)



Liseiki & Raymo (2004)

# GLACIAL ISOSTATIC ADJUSTMENT



# Some Observations Used in GIA Modelling

## Paleo

Relative sea level  
(shoreline deformation)

Lake levels (tilting)

Ice extent (geological  
archives)

## Geodetic

3D land motion (GPS)

Gravity changes (in-situ, GRACE)

Earth rotation changes (True Polar  
Wander, Length of Day).

Relative sea level (tide gauges,  
satellite altimetry)

Ice extent (satellite)

# GIA Model Components

## Earth Forcing

### Surface Loading

**Ice model**

**Sea-level model**

**Erosion/Deposition  
model**

### Rotational Potential

**Euler equations**

## Earth Model

### Geometry

**Spherical/Flat**

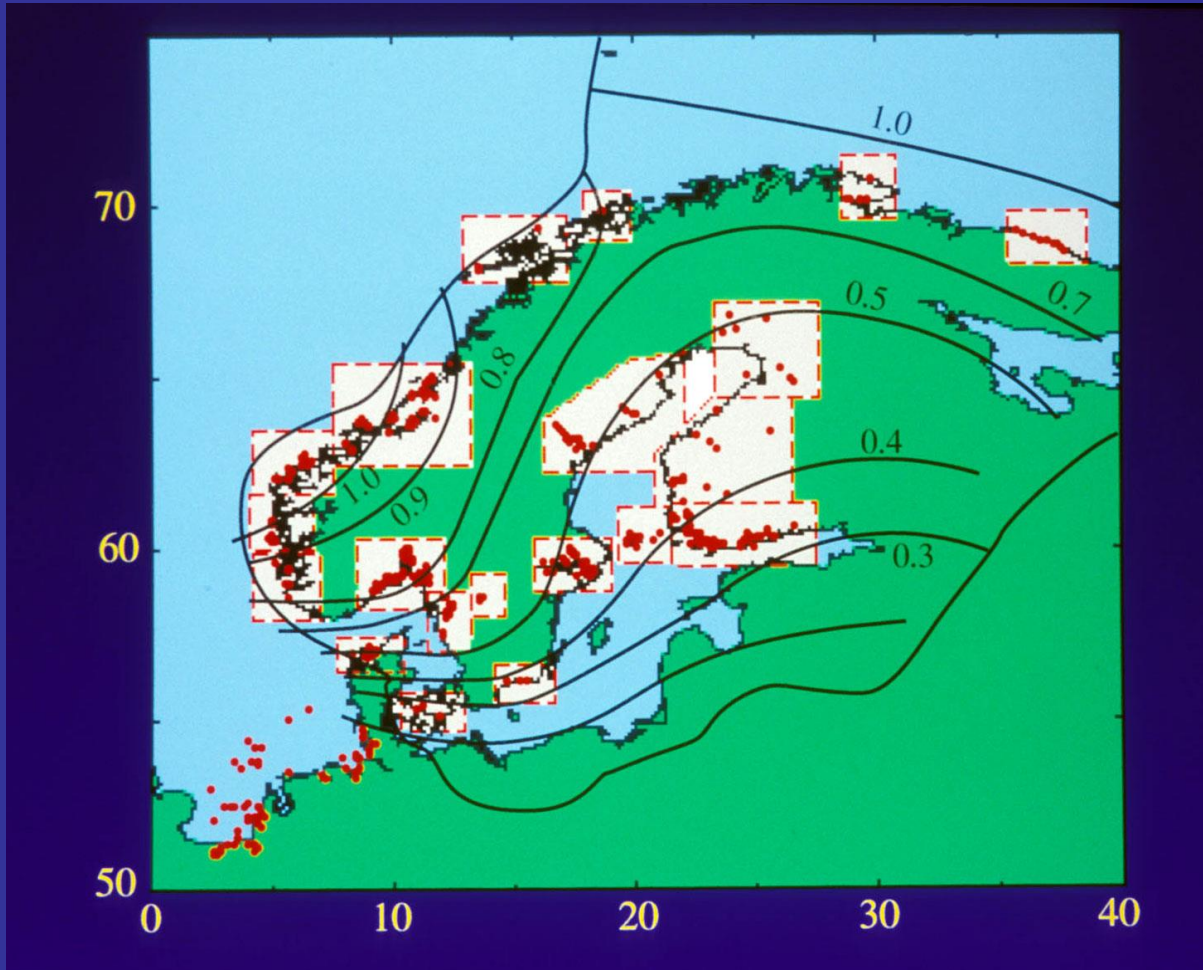
**Internal structure: 1D  
& 3D**

### Rheology

**Viscoelastic**

**Linear and non-  
linear viscous  
deformation**

# Regional Ice Sheet Reconstruction



Lambeck et al. (1998)

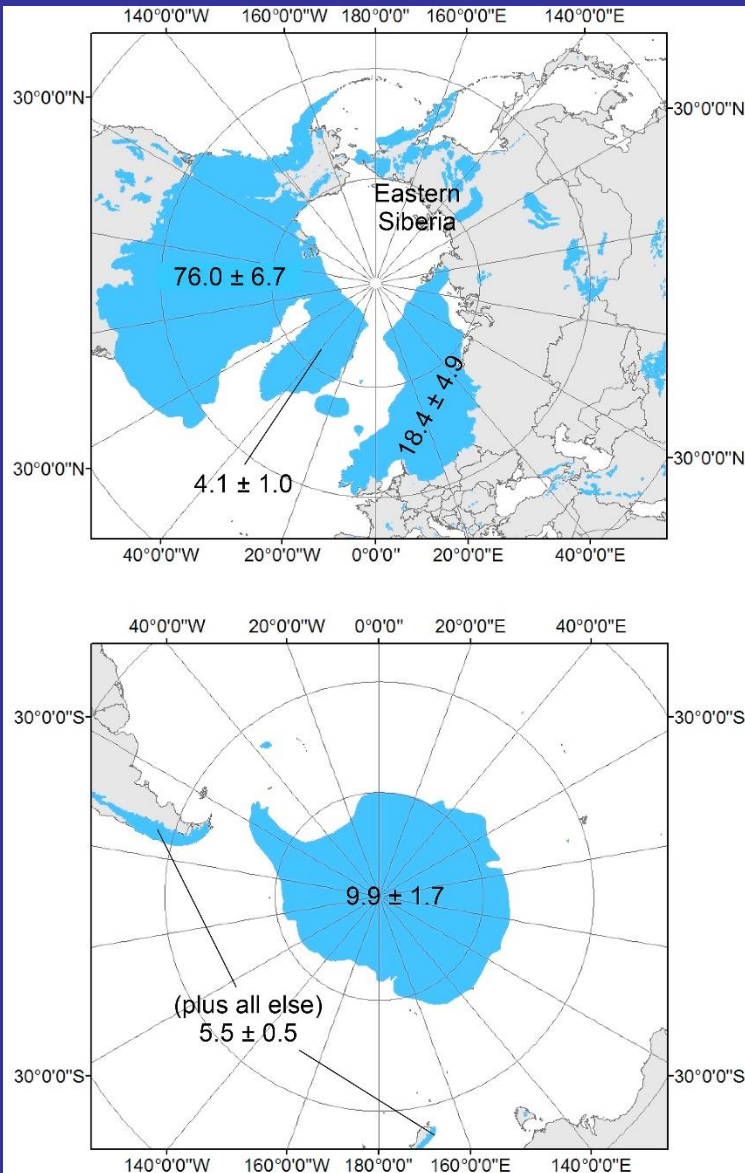
Direct data control from glacial geology and marine geophysics.

Indirect data control from GIA response (e.g. RSL, GPS).

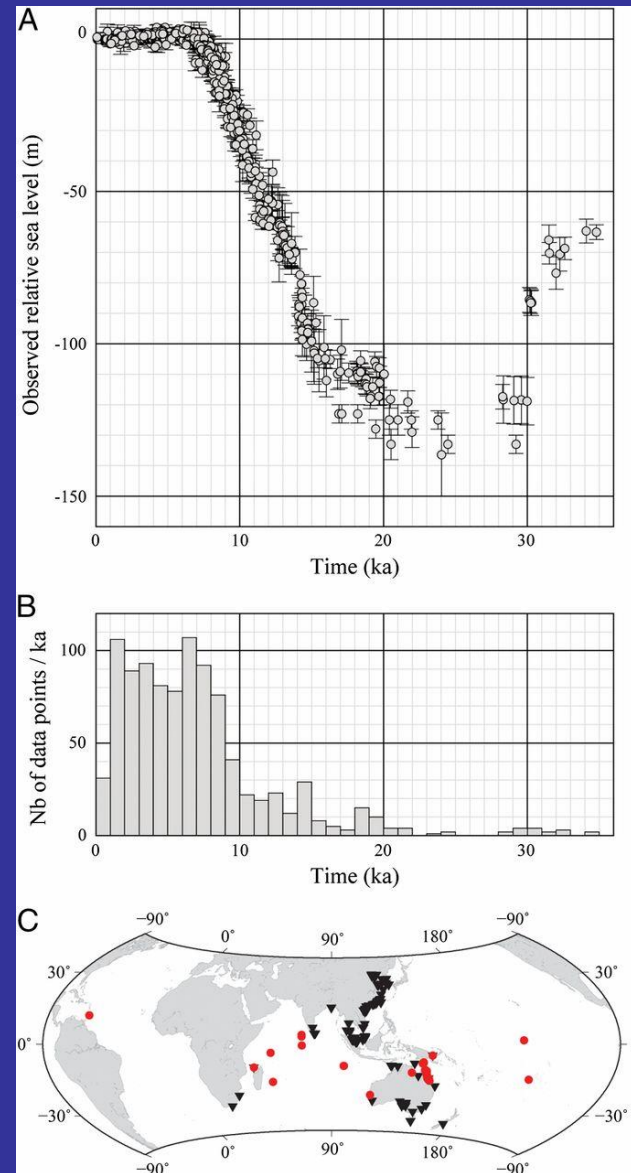
Constraints from glaciology (e.g. ice height profile, glaciological models).

For contemporary ice sheets can also use information from ice core records.

# Global Ice Sheet Reconstruction



Simms et al. (2019)



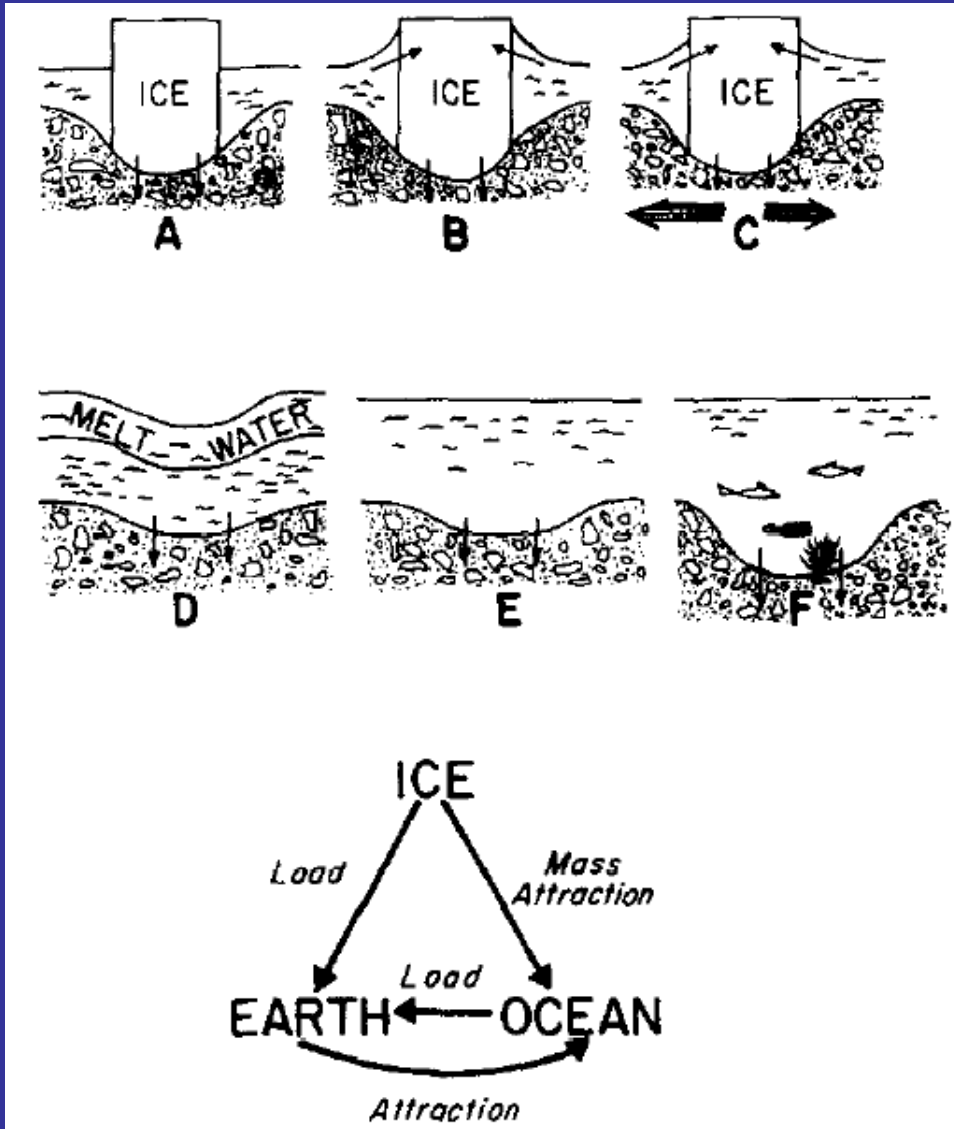
Lambeck et al. (2014)

Determined by summing regional reconstructions.

Volume of model reconstruction can be compared to that inferred from far-field RSL reconstructions.

At present these independent volume estimates do not compare well (see Clark and Tarasov, 2014).

# Sea-Level Model



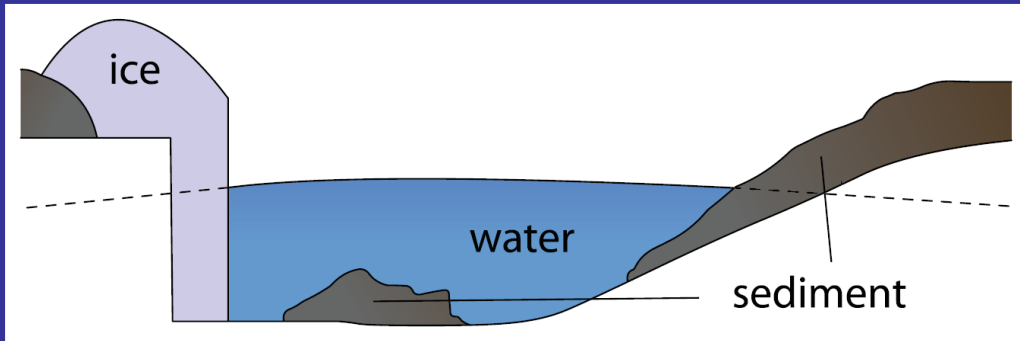
Compute sea-level change for a given global ice reconstruction and Earth model using the sea-level equation (Farrell and Clark, 1976).

Solutions of the sea-level eqn give model estimates of sea-level change as well as the ocean load.

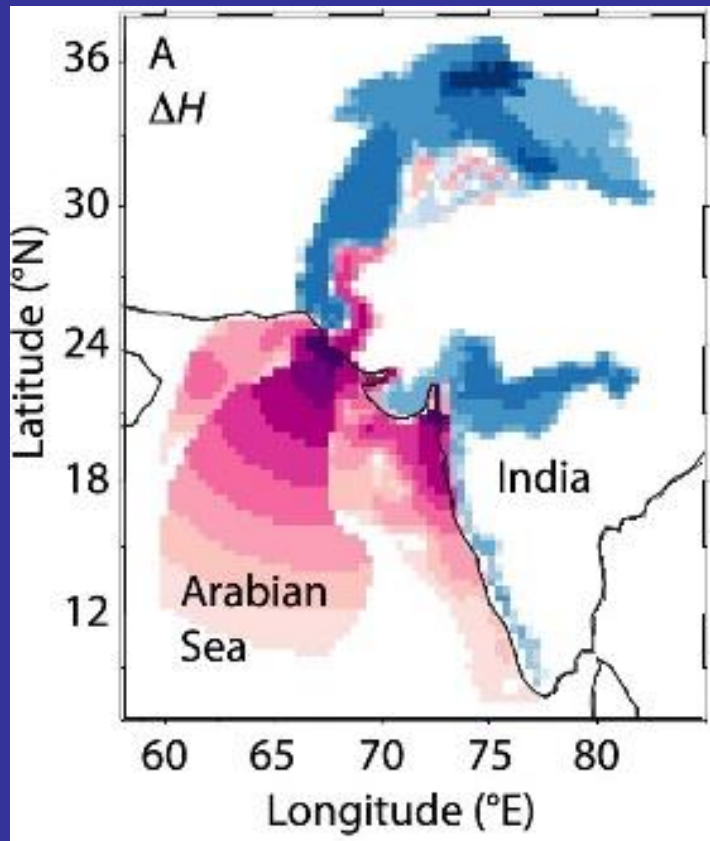
See Milne and Shennan (2013) for a review.

# Erosion/Deposition Model

Dalca et al. (2013)



Sediment erosion/deposition influences sea-level and so including a sediment load requires that an extended sea-level equation be solved (Dalca et al, 2013).



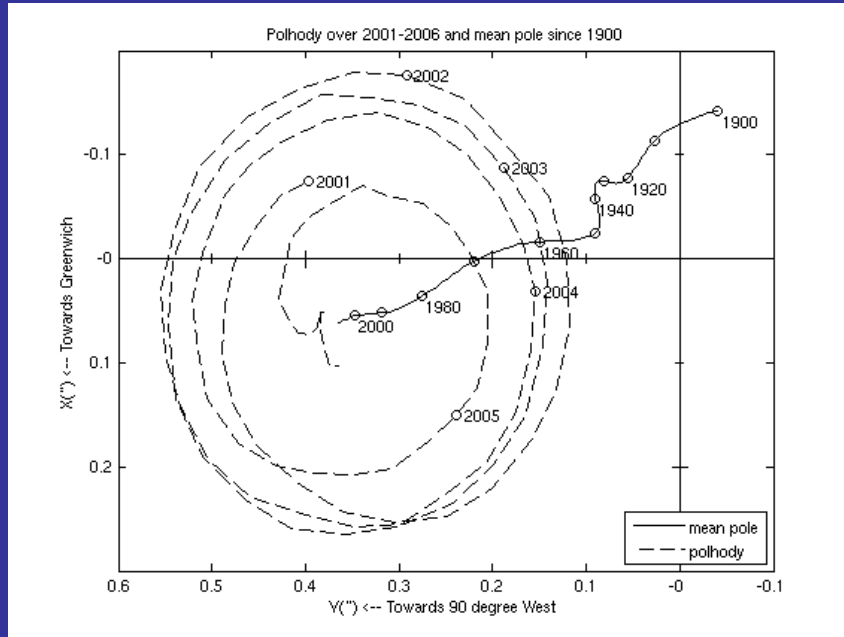
Influence of sediment loading most substantial near large river deltas (e.g. Arabian Sea, US Gulf coast – Kuchar et al. (2018)).

Since this load is cumulative over time its impact is greater for older times (e.g. Pliocene – Moucha et al. (2017)).

Ferrier et al. (2015)

# Changes in Earth Rotation

[www.iers.org/MainDisp.csl?pid=95-89](http://www.iers.org/MainDisp.csl?pid=95-89)

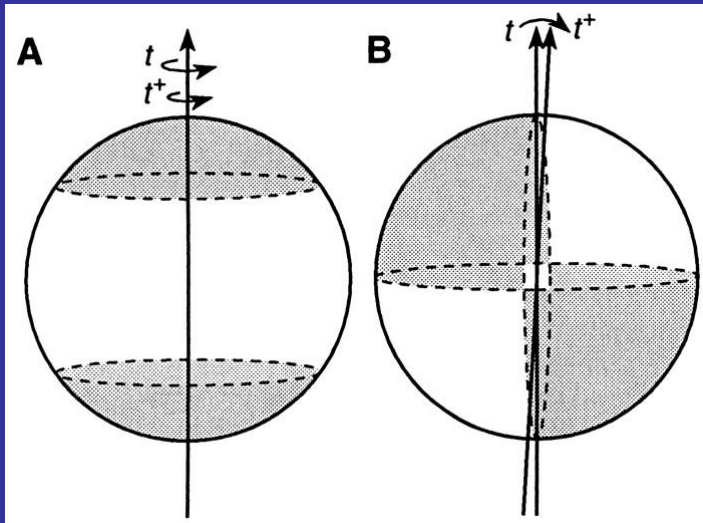


GIA changes the Earth's rotation vector (True Polar Wander and Length of Day).

Measurements of these changes can be used to constrain GIA models.

These changes also act as a secondary forcing via the 'pole tide' effect.

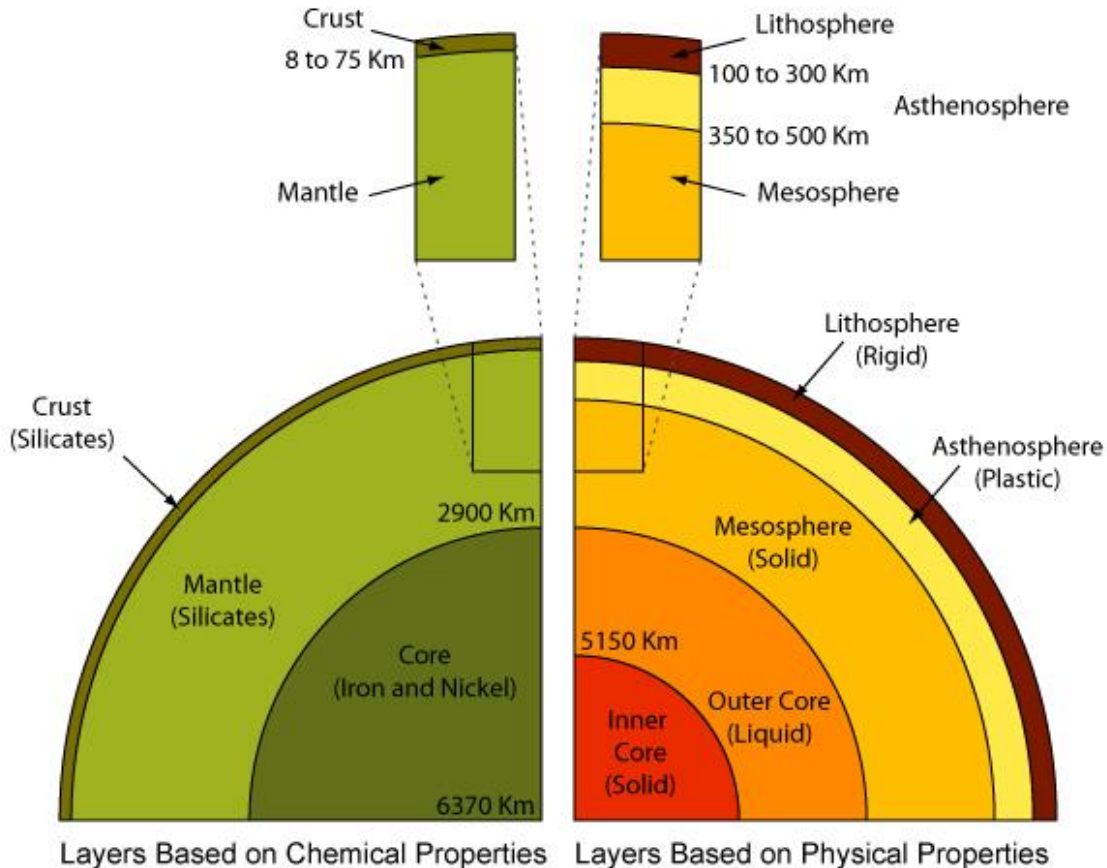
See Mitrovica and Wahr (2011).



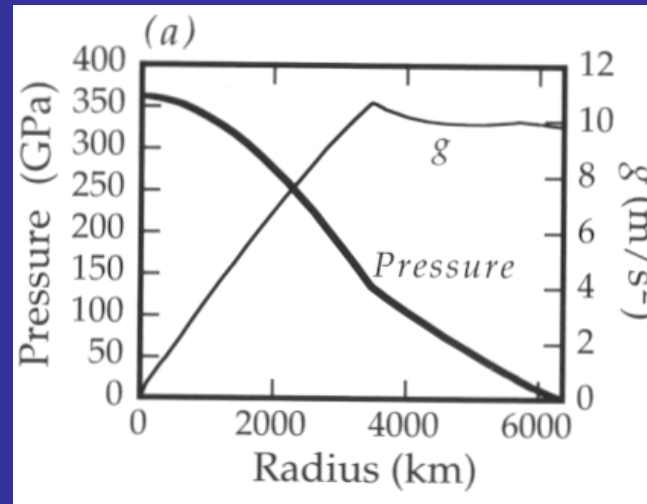
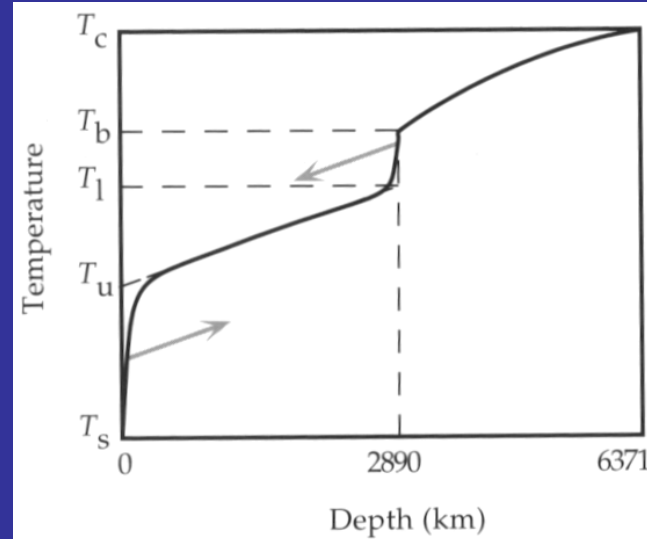
Mitrovica et al. (2001)

# Earth Model (I-D)

## CHEMICAL AND PHYSICAL LAYERS OF THE EARTH



Enduring Resources for Earth Science Education — <http://earthref.org/ERESE>  
<http://earthref.org/cgi-bin/erda.cgi?n=422>



Davies (1999)

Density and elastic properties from seismic models.

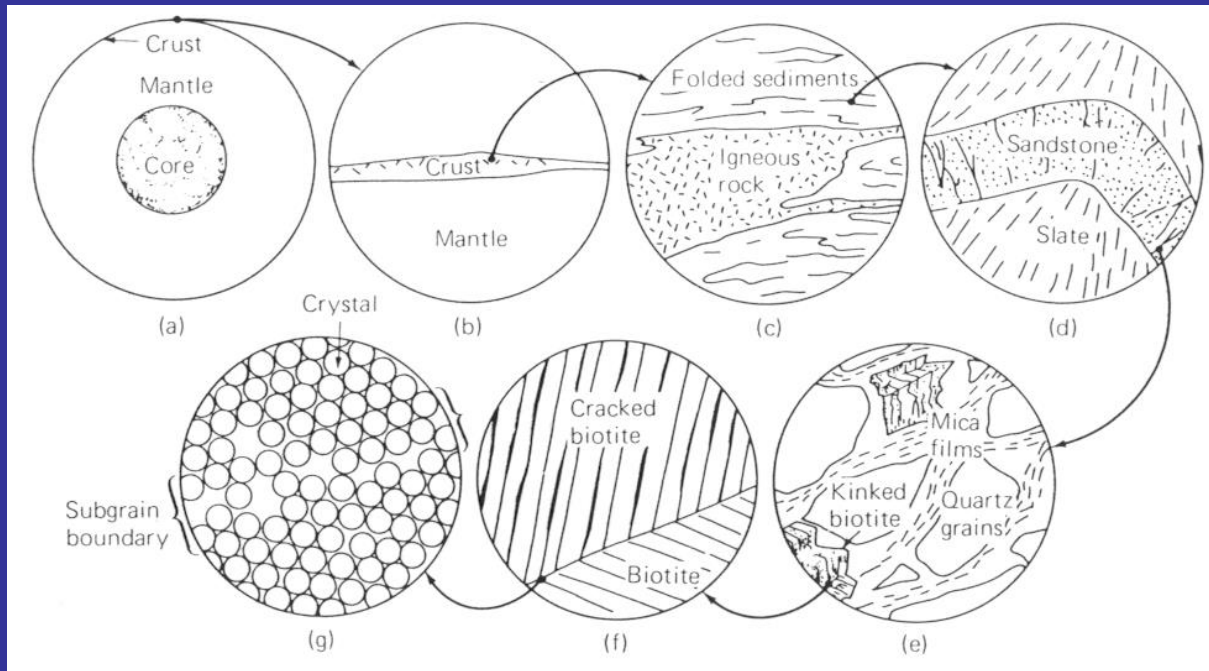
Temperature and pressure are the dominant external controls on rock viscosity.

Viscosity structure can be estimated via results from lab deformation experiments or geophysical modelling.

# Earth Model: Rheology

Bulk response governed by processes at grain and sub-grain scale.

Which processes are active at a given time depends on a variety of factors.



Means (2012)

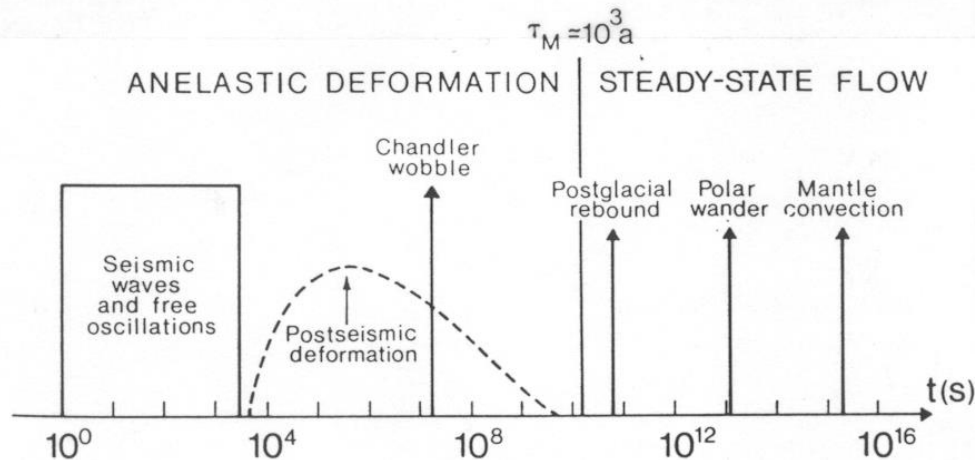
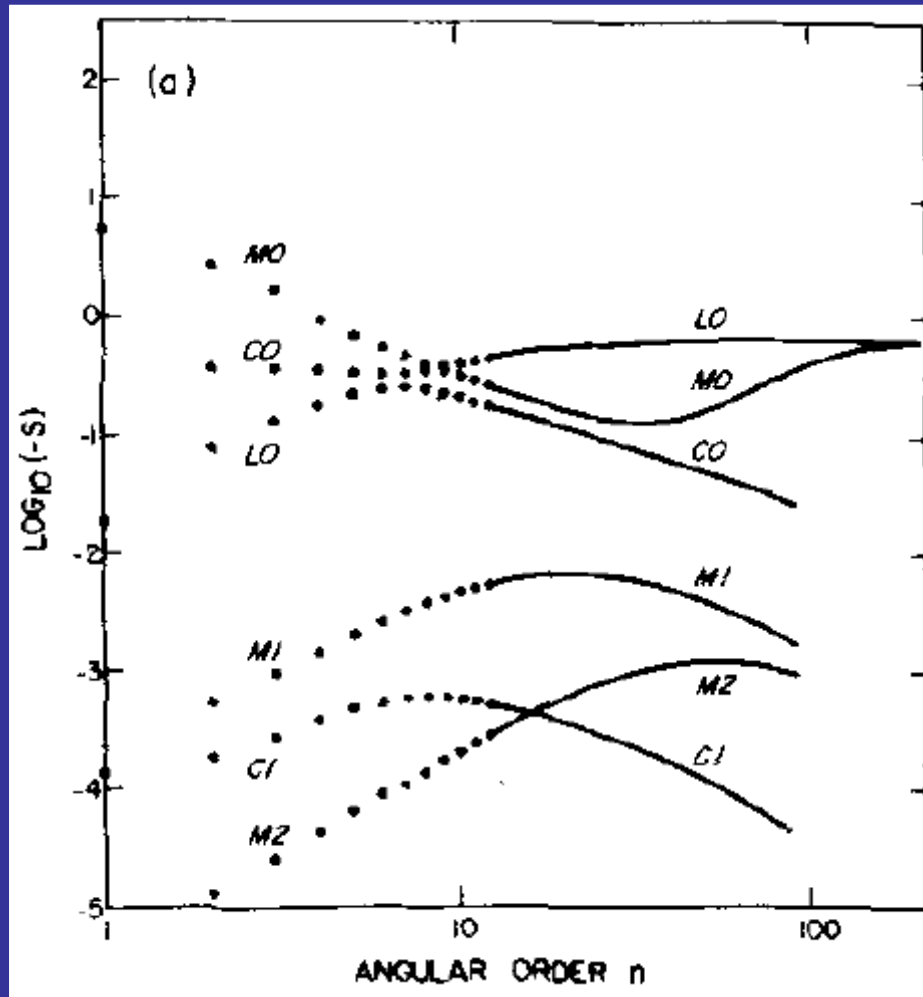


Figure 8.4 Characteristic timescales of mantle deformation processes. The Maxwell time is denoted by  $\tau_M$  (from Yuen *et al.* 1982).

Maxwell material is the 'default' rheological model for GIA but more complex rheologies have been explored (non-linear creep, transient creep, brittle failure)

Ranalli (1995)

# Normal Mode Formalism for Spherically Symmetric, Self-Gravitating, Visco-Elastic Earth Models



Consider the response of Earth model to an impulse forcing (Peltier, 1974).

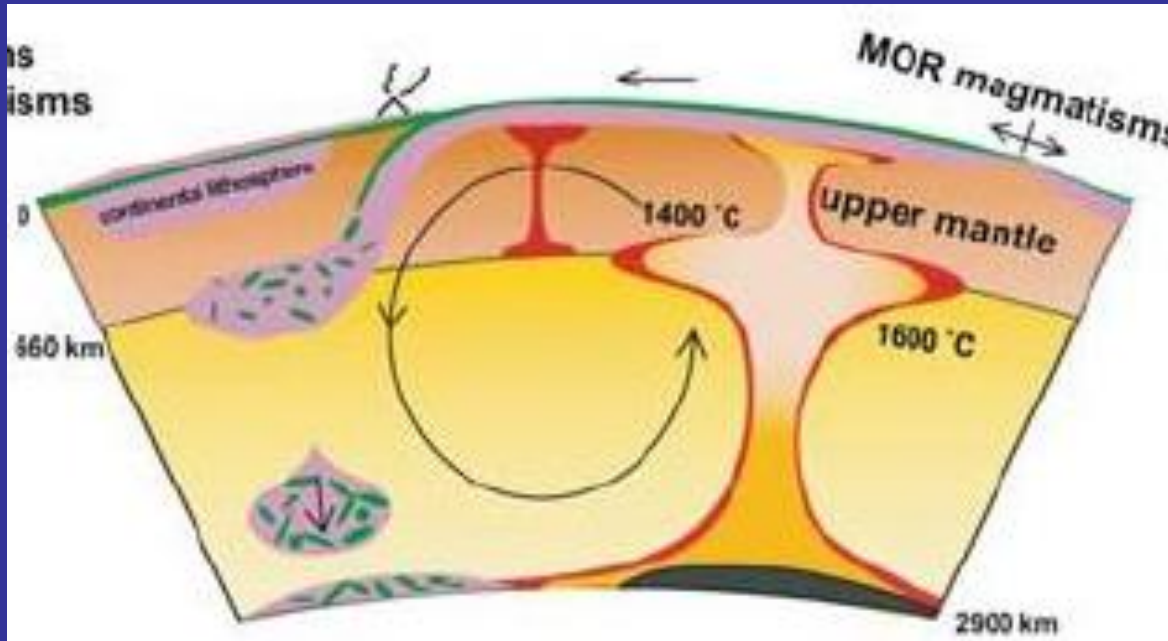
Simultaneously solve three equations: (i) momentum equation, (ii) continuity equation, (iii) Poisson's equation. Response of model to an impulse forcing.

Solution expressed in terms of Love numbers.

Peltier (1974), Wu & Peltier (1982)

# 3-D Earth Models

Santosh et al. (2010)



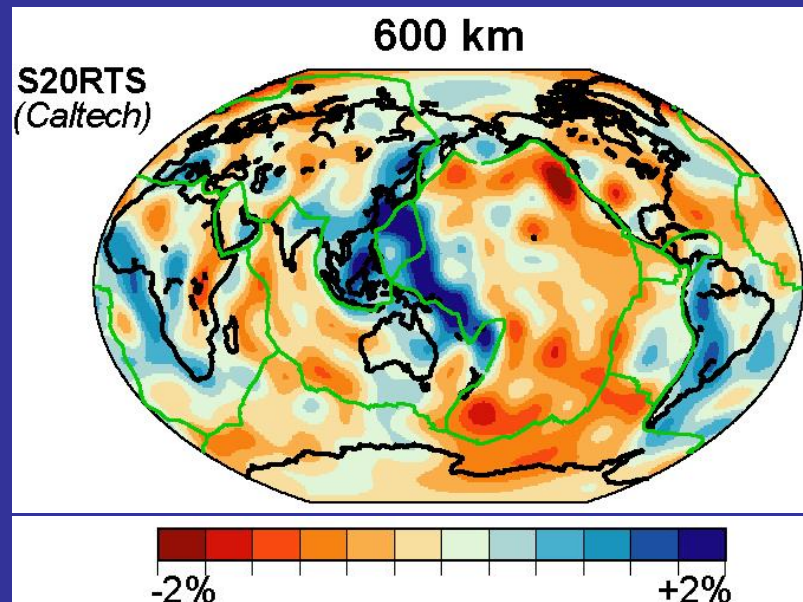
Lateral variations in Earth structure exist due to long-term Earth cooling.

These variations influence the rheology and therefore the GIA response.

There are a small number of groups that have developed numerical models that explicitly accommodate 3D Earth structure.

Defining lateral structure relies on using information from other geophysical methods (e.g. seismology).

Ritsema and van Heijst (2004)



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